the painting can be done as RBG, BRG, GRB...etc. But here it is possible that we may miss out some patterns of colours. So, we can use systematic listing as given below.

Step 1: Fix one colour; try the possible arrangements with other colours. For example, if the Third floor is fixed as Red, then we get 2 ways which is shown in the table below.

Fix Third floor Colour	Second floor Colour	First floor Colour
Red	Blue	Green
Red	Green	Blue

Step 2: If the Third floor is fixed as Blue, then we get another 2 ways which is shown in the table below.

Fix Third floor Colour	Second floor Colour	First floor Colour
Blue	Green	Red
Blue	Red	Green

Step 3: If the Third floor is fixed as Green, then we get 2 more ways which is shown in the table below.

Fix Third floor Colour	Second floor Colour	First floor Colour
Green	Blue	Red
Green	Red	Blue

Hence, we get 6 different ways of painting the three floors, which are R-B-G, R-G-B, B-G-R, B-R-G, G-B-R and G-R-B.

Situation 2

Suppose you want to write four digit numbers by using the digits 3, 6, 9 and 5. What are the possible numbers you can write using each digit exactly once?

If you list randomly, for example 9365, 3695, 5639 and so on, you may not write all the possibilities. So, write in ascending order.

- All numbers beginning with 3:

Fix next digit and change the other 2 digits. We get, 3569, 3596, 3659, 3695, 3956, 3965.

Similarly,

- All numbers beginning with 5 : 5369, 5396, 5639, 5693, 5936, 5963
- All numbers beginning with 6 : 6359, 6395, 6539, 6593, 6935, 6953
- All numbers beginning with 9 : 9356, 9365, 9536, 9563, 9635, 9653

  Totally, we can have 6+6+6+6 = 24 numbers

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